



Indian Institute of Information Technology, Surat  
Mid Semester Examination- 2025

B.Tech. (3rd Sem)

AS305 – Probability and Statistical Analysis

Date: 26/09/2025

Time Allowed: 90 Minutes

Max. Marks: 30

Instructions:

- (1) Write your Roll Number clearly on your answer book as well as every supplement you may use.
- (2) Answer all the questions.

1. (a) Find the mean of binomial distribution  $B(n, p)$  from its M.G.F. (moment generating function). (2)  
(b) A student attempts 2500 Codeforces problems. For each problem, the probability that the student solves it correctly is 0.002, independent of other problems. Find the probability that the student solves at least two problems in total. (2)  
(c) Let  $X$  be a Poisson random variable with variance  $1/2$ . Then find  $E[X + 1]!$ . (2)

2. (a) Let  $X$  be a random variable with p.d.f.

$$f(x) = \begin{cases} \frac{1}{x^2}, & \text{if } 1 < x < \infty; \\ 0 & \text{otherwise.} \end{cases}$$

Show that  $E[X]$  does not exist. (3)

- (b) Show that the function  $F_X : \mathbb{R} \rightarrow \mathbb{R}$  defined by

$$F_X(x) = \begin{cases} 0, & x < 0; \\ 1 - e^{-x}, & \text{if } x \geq 0. \end{cases}$$

is a distribution function. Further compute the probability density function. (3)

3. (a) At the IIIT Surat canteen, historically, 5% of the meals served generate a valid complaint about spoilage. The canteen committee inspects meals one-by-one. What is the probability that they must inspect at least 5 meals before they record 3 complaints? (3)  
(b) Suppose that the random variable  $X$  has the moment generating function

$$M_X(t) = c \sum_{k=-2}^2 \frac{e^{kt}}{k^2 + 1}, \quad -\infty < t < \infty$$

where  $c$  is a real constant. Then

- (i) find the value of  $c$ .
- (ii) Derive the probability density/mass function of  $Y = X^2 + |X|$ .
- (iii) Compute  $E[Y]$  and  $\text{Var}[Y]$ .

(1+2+2)

4. An economist observes that the relationship between monthly income ( $X$  in thousands) and monthly expenditure ( $Y$  in thousands) of families may not be perfectly linear. He decides to fit a second-degree polynomial regression model of the form  $\hat{Y} = a + bX + cX^2$ . If the data collected from 7 families is as follows:

| Family      | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|
| Income      | 10 | 15 | 20 | 25 | 30 | 35 | 40 |
| Expenditure | 8  | 12 | 19 | 27 | 36 | 47 | 60 |

Then:

Please turn over...

- (i) Fit a second-degree polynomial regression model to the above data. (2)
- (ii) Use the fitted model to predict the expenditure when income is 28 thousand. (1)
- (iii) Compute the Mean Squared Error (MSE) of the quadratic regression model. (2)

5. A company records the following data for 7 months: Advertisement Spending (X, in lakhs), Product Price (Y, in Rs per unit), and Sales (Z, in 1000 units).

| Month         | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|---------------|----|----|----|----|----|----|----|
| Advertisement | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
| Price         | 50 | 48 | 47 | 46 | 44 | 43 | 42 |
| Sales         | 40 | 42 | 50 | 55 | 60 | 65 | 70 |

then find:

- (i) the covariance between advertisement spending and sales. (1)
- (ii) the correlation coefficient between product price and sales. (2)
- (iii) the partial correlation between advertisement spending and sales, controlling for product price. (2)

**\*Best wishes\***